NUMBER SERIES

- Missing Number Series
- Wrong Number Series

5, 10, 20, 35, 55, 80 → Series

+5, +10, +15, +20, +25 → Pattern

Patterns

- Difference Pattern
- Multiplication or Division Pattern
- Miscellaneous Pattern.

★ Missing Number Series :—

⇒ Difference

$10 \xrightarrow{+7} 17 \xrightarrow{+14} 31 \xrightarrow{+21} 52 \xrightarrow{+28} 80 \xrightarrow{+35} 115$

⇒ Multiplication

$10 \xrightarrow{\times 2} 20 \xrightarrow{\times 4} 80 \xrightarrow{\times 6} 480 \xrightarrow{\times 8} 3840$

Pattern 1 :— Constant Difference (Arithmetic Progression)

Rule :- Add / Subtract a fix number (Common Difference)

Example,

•) $9 \xrightarrow{+5} 14 \xrightarrow{+5} 19 \xrightarrow{+5} 24 \xrightarrow{+5} 29$

•) $50 \xrightarrow{-5} 45 \xrightarrow{-5} 40 \xrightarrow{-5} 35 \xrightarrow{-5} 30$

Practice Questions:

Q1. 45, 40, 35, 30, 25, ?

(A) 18 (B) 17 (C) 20 (D) 15 (E) 24

$45 \xrightarrow{-5} 40 \xrightarrow{-5} 35 \xrightarrow{-5} 30 \xrightarrow{-5} 25 \xrightarrow{-5} ? \rightarrow 20$

Q2. 63, 55, 47, 39, ?, 23

(A) 31(B)33(C)35(D)29(E)None of these

$63 \xrightarrow{-8} 55 \xrightarrow{-8} 47 \xrightarrow{-8} 39 \xrightarrow{-8} ? \xrightarrow{-8} 23$

$39-8 = 31$

Q3. ?, 47, 54, 61, 68, 75

(A) 39 (B) 40 (C)36 (D)42 (E)41

$? \xrightarrow{+7} 47 \xrightarrow{+7} 54 \xrightarrow{+7} 61 \xrightarrow{+7} 68 \xrightarrow{+7} 75$

$x+7 = 47$

$x = 47-7$

$x = 40$

Q4. 116, 110, 104, 98, 92, ?

(A) 86 (B) 85 (C) 96 (D) 91 (E) 81

$116 \xrightarrow{-6} 110 \xrightarrow{-6} 104 \xrightarrow{-6} 98 \xrightarrow{-6} 92 \xrightarrow{-6} ? \rightarrow 92-6 = 86$

Q5. 96, 105, 114, 123, ?, 141

(A) 134 (B)135 (C)128 (D)132 (E)133

$96 \xrightarrow{+9} 105 \xrightarrow{+9} 114 \xrightarrow{+9} 123 \xrightarrow{+9} ? \xrightarrow{+9} 141$

$123+9 = 132$

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Pattern 2 → Double Difference (Increasing Order)

Rule :- Difference between terms keep increasing in a pattern.

Example,

- 6, 7, 9, 12, 16, 21

  +1 +2 +3 +4 +5

  +1 +1 +1 +1

- 3, 5, 9, 15, 23, 33

  +2 +4 +6 +8 +10

  +2 +2 +2 +2

Practice Questions:

Q1. 12, 17, 26, 39, 56, ?

(A) 89 (B) 81 (C) 77 (D) 84 (E) 65

12, 17, 26, 39, 56, ? → 56+21 = 77

+5 +9 +13 +17 +21

+4 +4 +4 +4

Q2. 7, 12, 18, 25, ?, 42

(A) 34 (B) 33 (C) 30 (D) 28 (E) 37

7, 12, 18, 25, ?, 42 → 25+8 = 33

+5 +6 +7 +8 +9

+1 +1 +1 +1

Q3. 16, 9, 6, 7, ?, 21

(A) 15 (B)8 (C)6 (D)12 (E)10

16, 9, 6, 7, ?, 21 → 7+5 = 12

−7 −3 +1 +5 +9

+4 +4 +4 +4

Pattern 3 → Double Difference (Decreasing Order)

Rule :- Differences between terms keep decreasing in a pattern.

Examples,

- 110, 90, 74, 62, 54, 50

  −20 −16 −12 −8 −4

  +4 +4 +4 +4

- 90, 61, 39, 24, 16, 15

  −29 −22 −15 −8 −1

  +7 +7 +7 +7

Practice Questions:

Q1. 125, 104, 87, 74, 65, ?

(A) 56 (B) 62 (C) 64 (D) 53 (E) 60

125, 104, 87, 74, 65, ? → 65 − 5 = 60

−21 −17 −13 −9 −5

+4 +4 +4 +4

Q2. ?, 132, 101, 79, 66, 62

(A) 174 (B) 165 (C) 163 (D) 170 (E) 172

$x - 40 = 132$

$x = 132 + 40$

$x = 172$

?, 132, 101, 79, 66, 62

−40 −31 −22 −13 −4

+9 +9 +9 +9

Q3.185, 174, 166, 161, 159, ?
(A) 150 (B)157 (C)154 ~~(D)~~ 160 (E)155

185, 174, 166, 161, 159, ? → 159 + 1 = 160

−11 −8 −5 −2 +1

+3 +3 +3 +3

Pattern 4 :- Alternate Differences

Rule :- Two different differences repeat alternatively.

Patterns can be →
1. +a, +b, +a, +b, +a, +b, ----
2. −a, −b, −a, −b, −a, −b, -----
3. +a, −b, +a, −b, +a, −b, -----
4. −a, +b, −a, +b, −a, +b, -----

Examples,

- 4, 6, 11, 13, 18, 20, 25
  +2 +5 +2 +5 +2 +5
- 30, 27, 34, 31, 38, 35, ? → 42
  −3 +7 −3 +7 −3 +7
- 4, 6, 11, 13, 18, 20, 25, 27
  +2 +5 +2 +5 +2 +5 +2

Practice Questions:
Q1. 38, 43, 37, 42, 36, ?
(A) 42 ~~(B)~~ 41 (C) 36 (D) 40 (E) 38

38, 43, 37, 42, 36, ? → 36 + 5 = 41

+5 −6 +5 −6 +5

Q2. 48, 51, 57, 60, ?, 69

(A) 56 (B) 66 (C) 65 (D) 59 (E) 62

48, 51, 57, 60, ?, 69

+3 +6 +3 +6 +3

60 + 6 = 66

Q3. ?, 71, 69, 76, 74, 81

(A) 64 (B) 52 (C) 61 (D) 68 (E) 67

?, 71, 69, 76, 74, 81

+7 −2 +7 −2 +7

$x + 7 = 71$

$x = 71 - 7$

$x = 64$

Q4. 51, 54, 58, 61, 65, ?

(A) 73 (B) 70 (C) 61 (D) 71 (E) 68

51, 54, 58, 61, 65, ?

+3 +4 +3 +4 +3

65 + 3 = 68

Q5. 35, 33, 28, 26, 21, ?

(A) 16 (B)18 (C)20 (D)15 (E)19

35, 33, 28, 26, 21, ?

−2 −5 −2 −5 −2

21 − 2 = 19

Pattern 5 → Prime Differences

Rule :- Differences are consecutive Prime Numbers

Prime Numbers :-

A natural number > 1, with exactly two distinct positive factors → 1 and itself.

1 – 100 Prime Numbers

{ 2, 3, 5, 7, 11, 13, 17, 19, 23,
29, 31, 37, 41, 43, 47, 53, 59,
61, 67, 71, 73, 79, 83, 89, 97 }

Prime Numbers

1 – 100 → 25 prime numbers
1 – 50 → 15 prime numbers
50 – 100 → 10 prime numbers

• **Definitions**

Even Numbers →

Numbers divisible by 2 are even Numbers. → like, 10, 12, 4, 6, 8, 10, 12, −2, −8, −48, −4720, −8292, ----

Odd Numbers →

Numbers that are not divisible by 2 are odd Numbers. → 3, 7, 9, 21, 43, −81, −433, −8295, ---

Prime Numbers →

Numbers that are divisible by 1 and itself are Prime Numbers. → 2, 3, 5, 7, 11, 13, 17, 19, ---

Examples,

- 4, 7, 12, 19, 30, 43, 60

  +3 +5 +7 +11 +13 +17

  {Consecutive Prime Numbers}

- 10, 12, 15, 20, 27, 38, 51

  +2 +3 +5 +7 +11 +13

Practice Questions:

Q1. 13, 72, 133, 200, 271, 344, ?

(A) 424 (B) 423 (C) 418 (D) 398 (E) 422

13, 72, 133, 200, 271, 344, ?

+59 +61 +67 +71 +73 +79

344 + 79 = 423

Q2. 13, 32, 55, 84, 115, ?, 193

(A) 147 (B) 153 (C) 152 (D) 145 (E) 151

13, 32, 55, 84, 115, ?, 193

+19 +23 +29 +31 +37 +41

115 + 37 = 152

Q3. ?, 30, 43, 60, 79, 102, 131

(A) 24 (B) 19 (C) 14 (D) 22 (E) 20

?, 30, 43, 60, 79, 102, 131

+11 +13 +17 +19 +23 +29

$x + 11 = 30$

$x = 30 - 11$

$x = 19$

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Pattern 6 → Square Differences

Rule :- Differences are consecutive square numbers (1, 4, 9, 16, 25, ----)

Examples,

- 2, 3, 7, 16, 32, 57

  +1 (1^2), +4 (2^2), +9 (3^2), +16 (4^2), +25 (5^2)

- 10, 14, 23, 29, 64, 100

  +4 (2^2), +9 (3^2), +16 (4^2), +25 (5^2), +36 (6^2)

Practice Questions:

Q1. 19, 23, 32, 48, 73, ?

(A) 124 (B) 111 (C) 110 (D) 109 (E) 112

19, 23, 32, 48, 73, ?

+4, +9, +16, +25, +36

73+36 = 109

Q2. 194, ?, 81, 45, 20, 4

(A) 131 (B) 132 (C) 137 (D) 134 (E) 130

194, ?, 81, 45, 20, 4

−64 (8^2), −49 (7^2), −36 (6^2), −25 (5^2), −16 (4^2)

194 − 64 = 130

Q3. 120, 156, 220, 320, 464, 660

(A) 448 (B) 452 (C) 460 (D) 464 (E) 468

(F) 104

?, 120, 156, 220, 320, 464, 660

+16 (4^2), +36 (6^2), +64 (8^2), +100 (10^2), +144 (12^2), +196 (14^2)

$x + 16 = 120$

$x = 120 - 16$

$x = 104$

Pattern 7 :– Cube Differences

Rule :– Differences are consecutive cube Numbers.
(1, 8, 27, 64, 125, ---)

Examples,

- 3, 4, 12, 39, 103, 228

  $3 \xrightarrow{+1} 4 \xrightarrow{+8} 12 \xrightarrow{+27} 39 \xrightarrow{+64} 103 \xrightarrow{+125} 228$

  (1^3) (2^3) (3^3) (4^3) (5^3)

- 5, 13, 40, 104, 229, 445

  $5 \xrightarrow{+8} 13 \xrightarrow{+27} 40 \xrightarrow{+64} 104 \xrightarrow{+125} 229 \xrightarrow{+216} 445$

  (2^3) (3^3) (4^3) (5^3) (6^3)

Practice Questions:

Q1. 19, 27, 54, 118, 243, ?

(A) 456 (B) 459 (C) 471 (D) 439 (E) 460

$19 \xrightarrow{+8} 27 \xrightarrow{+27} 54 \xrightarrow{+64} 118 \xrightarrow{+125} 243 \xrightarrow{+216} ?$

(2^3) (3^3) (4^3) (5^3) (6^3)

243 + 216
= 459

Q2. 2000, 1488, 1145, 929, 804, ?

(A) 710 (B) 740 (C) 720 (D) 750 (E) 730

$2000 \xrightarrow{-512} 1488 \xrightarrow{-343} 1145 \xrightarrow{-216} 929 \xrightarrow{-125} 804 \xrightarrow{-64} ?$

(8^3) (7^3) (6^3) (5^3) (4^3)

804 − 64
= 740

Q3. 50, 58, 122, 338, ?, 1850

(A) 720 (B)850 (C) 1040 (D)790 (E)910

$50 \xrightarrow{+8} 58 \xrightarrow{+64} 122 \xrightarrow{+216} 338 \xrightarrow{+512} ? \xrightarrow{+1000} 1850$

(2^3) (4^3) (6^3) (8^3) (10^3)

338 + 512
= 850

Pattern 8 → Alternate Square Difference

Rule :- Square differences with alternate sign (+|−)

Examples,

- 20, 24, 15, 31, 6, 42

  +4 (2^2), −9 (3^2), +16 (4^2), −25 (5^2), +36 (6^2)

- 40, 49, 33, 58, 22, 71

  +9 (3^2), −16 (4^2), +25 (5^2), −36 (6^2), +49 (7^2)

Practice Questions:

Q1. 120, ?, 113, 138, 102, 151

(A) 129 (B) 124 (C) 136 (D) 121 (E) 125

120, ?, 113, 138, 102, 151

+9, −16, +25, −36, +49

$+(3^2)$ $-(4^2)$ $+(5^2)$ $-(6^2)$ $+(7^2)$

120 + 9
= 129

Q2. 300, ?, 275, 471, 246, 502

(A) 452 (B) 456 (C) 441 (D) 444 (E) 463

300, ?, 275, 471, 246, 502

+144, −169, +196, −225, +256

(12^2) $-(13^2)$ $+(14^2)$ $-(15^2)$ $+(16^2)$

300 + 144
= 444

Q3. 200, ?, 184, 233, 152, 273

(A) 207 (B) 209 (C) 211 (D) 213 (E) 205

200, ?, 184, 233, 152, 273

+9, −25, +49, −81, +121

$+(3^2)$ $-(5^2)$ $+(7^2)$ $-(9^2)$ $+(11^2)$

200 + 9
= 209

Pattern 9 → Alternate Cube Difference

Rule :- Cube Difference with Alternate Sign (+|-)

Example,

- 50, 51, 43, 70, 6, ?

  +1 -8 +27 -64 +125

  $+(1^3)$ $-(2^3)$ $+(3^3)$ $-(4^3)$ $+(5^3)$

- 120, 128, 101, 165, 40, 256

  +8 27 +64 -125 +216

  $+(2^3)$ $-(3^3)$ $+(4^3)$ $-(5^3)$ $+(6^3)$

Practice Questions:

Q1. 300, ?, 281, 345, 220, 436

(A) 325 (B) 312 (C) 316 (D)292 (E) 308

300, ?, 281, 345, 220, 436

+8 -27 +64 -125 +216

(2^3) (3^3) (4^3) (5^3) (6^3)

300 + 8 = 308

Q2. 1000, ?, 909, 1252, 740, 1469

(A) 1008 (B) 1064 (C) 1089 (D)1125 (E) 1144

1000, ?, 909, 1252, 740, 1469

+125 -216 +343 -512 +729

(5^3) (6^3) (7^3) (8^3) (9^3)

1000 + 125 = 1125

Q3. 500, 527, 402, 745, 16, ?

(A) 1260 (B)1347 (C)1296 (D)1300 (E)1325

500, 527, 402, 745, 16, ?

+27 -125 +343 -729 +1331

(3^3) (5^3) (7^3) (9^3) (11^3)

16 + 1331 = 1347

Pattern 10 :- Cubic Series

Rule :- Based on n^3 or $n^3 \pm n$ formulas.

Examples,

$(n^3+n) \rightarrow$ 2, 10, 30, 68, 130, 222, ---

$(1+1)$, $(8+2)$, $(27+3)$, $(64+4)$, $(125+5)$, $(216+6)$

(1^3+1), (2^3+2), (3^3+3), (4^3+4), (5^3+5), (6^3+6)

$n(n+1) \longrightarrow$ 1×2, 2×3, 3×4, 4×5, 5×6, 6×7 ---

2, 6, 12, 20, 30, 42

Practice Questions:

Q1. 28, 65, 126, 217, ?, 513
(A) 384 (B) 344 (C) 351 (D) 342 (E) 337

28, 65, 126, 217, ?, 513

(3^3+1), (4^3+1), (5^3+1), (6^3+1), (7^3+1), (8^3+1)

$\hookrightarrow 343+1 = 344$

Q2. 6, 13, 32, 69, 130, ?
(A) 221 (B) 196 (C) 261 (D) 217 (E) 233

6, 13, 32, 69, 130, ? $\rightarrow$ 221

$(1+5)$, $(8+5)$, $(27+5)$, $(64+5)$, $(125+5)$, $(216+5)$

(1^3+5), (2^3+5), (3^3+5), (4^3+5), (5^3+5), (6^3+5)